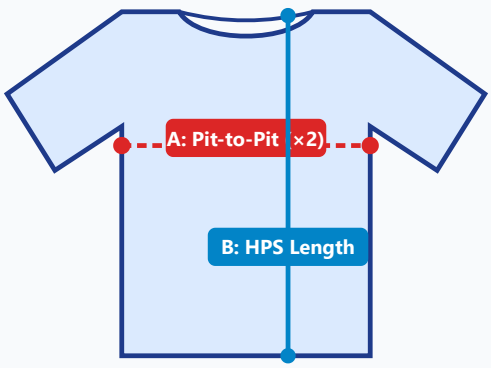
APPAREL ENGINEERING REFERENCE #1 · S–5XL

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How to Measure Chest Width & Body Length on a Flat Unisex T-Shirt

Pit-to-Pit Half-Chest vs. Body Chest Circumference + US Standard S–5XL Blank Specifications (Gildan 5000 vs. Bella+Canvas 3001)

A FLAT GARMENT MEASUREMENT DIAGRAM



Golden Rule: Garment Chest = **Flat Pit-to-Pit (A) × 2**. Measure 1 inch below armhole seams on a flat surface.

B BODY CHEST VS. GARMENT EASE FORMULA

Never confuse **Body Chest Circumference** (measured around your skin) with **Flat Shirt Width**. A proper fit requires positive breathing room (*design + fit ease*):

```
Garment Chest = (Flat Width × 2)
Target Shirt ≥ Body Chest + Fit Ease
```

Fit Preference	Chest Ease Added	Typical Feel
Snug / Athletic	+1.0 to +2.0 in (+3–5 cm)	Fitted chest & sleeves
Regular Standard	**+3.0 to +4.0 in (+8–10 cm)**	**Classic everyday drape**
Relaxed / Street	+5.0 to +6.5 in (+13–16 cm)	Order 1 size up

C US ADULT UNISEX T-SHIRT MASTER CHART (S–5XL: FLAT WIDTH, FULL CHEST & BODY RANGE)

Size	Flat Pit-to-Pit (A)	Full Garment Chest (A×2)	HPS Length (B)	Fits Body Chest (Regular)	Gildan 5000 vs. Bella+Canvas 3001 Note
S	18.0 in (45.7 cm)	36.0 in (91.4 cm)	28.0 in (71.1 cm)	34–36 in (86–91 cm)	Identical flat width; BC3001 tapered side-seam
M	**20.0 in (50.8 cm)**	**40.0 in (101.6 cm)**	**29.0 in (73.7 cm)**	**38–40 in (96–102 cm)**	**Most popular women's relaxed / men's athletic**
L	**22.0 in (55.9 cm)**	**44.0 in (111.8 cm)**	**30.0 in (76.2 cm)**	**42–44 in (107–112 cm)**	**#1 US men's size (5'10"–6'0", 175–205 lb)**
XL	24.0 in (61.0 cm)	48.0 in (121.9 cm)	31.0 in (78.7 cm)	46–48 in (117–122 cm)	Broad shoulders or mid-section comfort
2XL	26.0 in (66.0 cm)	52.0 in (132.1 cm)	32.0 in (81.3 cm)	50–52 in (127–132 cm)	Extended unisex cut; +2 in chest per step
3XL	28.0 in (71.1 cm)	56.0 in (142.2 cm)	33.0 in (83.8 cm)	54–56 in (137–142 cm)	Big & Tall generous torso drape
4XL	30.0 in (76.2 cm)	60.0 in (152.4 cm)	34.0 in (86.4 cm)	58–60 in (147–152 cm)	Available in Gildan 5000 Classic Heavy Tee
5XL	32.0 in (81.3 cm)	64.0 in (162.6 cm)	35.0 in (88.9 cm)	62–64 in (157–163 cm)	Maximum extended range (64 in garment chest)

ANSUR II & NHANES Calibration Finding (n = 6,068 adults): Estimating shirt size from Height + Weight + BMI matches a tape-measured chest **74% of the time** and lands within one size **99.9% of the time**. When between two sizes on 100% cotton blanks, always **size up** to account for ±0.75 in factory tolerance and 3–4% first-wash shrinkage.

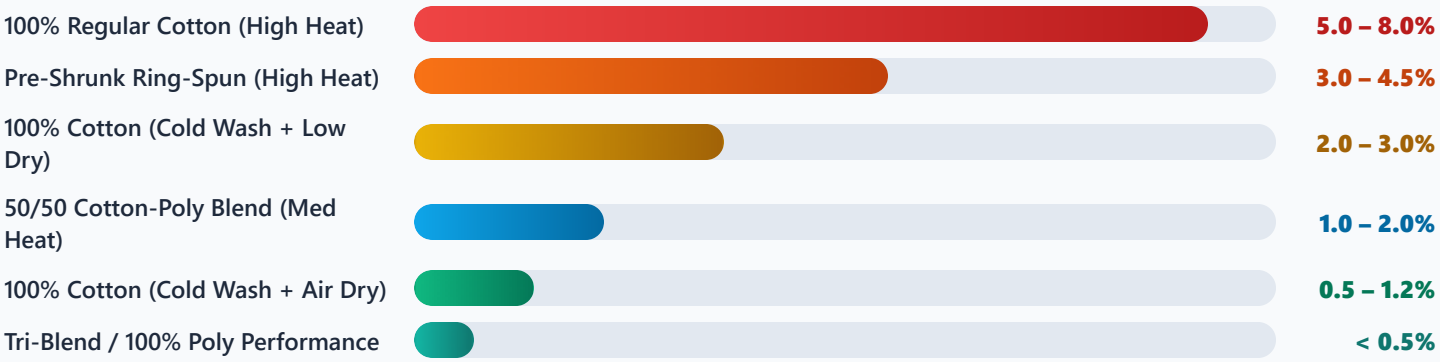
TEXTILE SHRINKAGE REFERENCE #2 · FABRIC PHYSICS

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100% Cotton vs. Ring-Spun vs. 50/50 Poly-Cotton Shrinkage Rate Chart

Dimensional Loss (%) & Real Inch Reduction on a Size Large Tee (22 × 30 in) Across Wash & Tumble-Dryer Heat Cycles

1 VISUAL SHRINKAGE RATE COMPARISON (HIGH-HEAT TUMBLE DRY VS. COLD WASH + AIR DRY)



2 EXACT DIMENSIONAL SHRINKAGE MATRIX BY FABRIC & LAUNDRY TEMPERATURE

Fabric Construction	Cold Wash + Air Dry	Warm Wash + Low Tumble	Hot Wash + High Dryer (150°F+)	Size Large (22×30") Loss on High Heat
100% Unwashed Loom Cotton	2.0 – 3.0%	4.5 – 6.0%	**6.0 – 9.0% (Severe)**	−1.3 in width / −2.2 in length (Drops 1 full size)
Pre-Shrunk Heavy Cotton (Gildan 5000)	**0.8 – 1.5%**	**2.5 – 3.5%**	**4.0 – 5.5%**	**−0.8 in width / −1.4 in length (Noticeably shorter)**
Combed Ring-Spun (Bella+Canvas 3001)	**0.5 – 1.2%**	**2.0 – 3.0%**	**3.5 – 4.5%**	**−0.7 in width / −1.2 in length (Snugger chest)**
50% Cotton / 50% Polyester Blend	0.2 – 0.5%	1.0 – 1.5%	**1.5 – 2.5%**	−0.3 in width / −0.6 in length (Holds shape)
Tri-Blend (50% Poly / 25% Cotton / 25% Rayon)	0.0 – 0.3%	0.5 – 1.0%	**1.0 – 1.5%**	Negligible (< 0.3 in change)
100% Micro-Polyester Performance Tee	0%	0 – 0.2%	**0.2 – 0.5% (Heat damage risk)**	Zero shrink, but high heat degrades wicking fibers

3 WHY LENGTH SHRINKS MORE THAN WIDTH

During circular jersey knitting, vertical **wale yarns (length)** are held under higher mechanical tension than horizontal **course loops (width)**. When hot water + tumble agitation relax cellulose hydrogen bonds, length contracts **1.5× to 1.8× more** than chest width.

4 WHAT "PRE-SHRUNK" ACTUALLY MEANS

Mill sanforization (compaction) removes initial loom tension, reducing potential shrinkage from ~9% down to **3–4% residual relaxation shrinkage**. It does *not* mean 0% shrink if baked in a 155°F commercial dryer.

Source & Full Testing Guide: tshirtsiwant.com/blog/does-cotton-shrink/

Size Up Calculator: tshirtsiwant.com/tools/size-calculator/

PRINT CONSERVATION PROTOCOL #3 · DTG & SCREEN PRINT

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The 5-Step Graphic Tee Care Protocol to Prevent Cracking & Peeling

How to Protect Water-Based DTG Ink & Plastisol Screen Prints Through 50+ Wash Cycles Without Fading or Collar Bacon

01 Turn the Shirt Inside-Out Before Every Wash & Dry

Why it works: Jeans zippers, metal buttons, and heavy towels act like sandpaper during drum agitation. Flipping the tee inside-out shields the cured ink layer from direct mechanical abrasion and prevents surface micro-scratches that turn into visible cracks.

02 Wash in Cold Water Only (≤ 85°F / 30°C) on Gentle Cycle

Why it works: DTG pigment binders and plastisol inks soften above 105°F (40°C) while cotton fibers swell and contract. Cold water prevents thermal shock between the elastic ink film and the underlying cotton yarn, cutting color fade by over 65%.

03 Use Mild Liquid Detergent — Zero Bleach, Zero Fabric Softener

Why it works: Chlorine/oxygen bleach oxidizes organic ink pigments, while liquid fabric softeners and dryer sheets coat fibers in silicone waxes that weaken ink-to-cotton adhesion and dissolve cured water-based pre-treatment polymers.

04 Wash With Similar Lightweight Fabrics & Low Spin (600–800 RPM)

Why it works: Washing lightweight graphic tees alongside heavy denim or fleece hoodies stretches wet rib-knit collars ("bacon neck") and twists side seams. Keep load weights balanced and avoid 1,200+ RPM high-extraction spins.

05 Air-Dry Flat or Hang from Bottom Hem (Or Tumble Dry Air-Only / Lowest Heat)

Why it works: High-heat tumble drying (140°F–160°F) is the #1 cause of cracked graphic tees: it bakes plasticizers out of the ink while shrinking the cotton beneath it. Hang via wide wooden/padded hangers inserted from the bottom hem—never shove a wire hanger through the neckband.

! QUICK TROUBLESHOOTING: WHY GRAPHIC PRINTS FAIL PREMATURELY

Observed Damage	Primary Root Cause in Laundry	Immediate Prevention Fix
Spiderweb Cracking Across Chest Print	High-heat dryer baking plasticizers + cotton shrinking underneath ink	Switch to Air Dry or Tumble Dry Low (<110°F); never iron directly on print
Faded / Chalky Vintage Look After 3 Washes	**Hot water + harsh enzyme/bleach detergent scrubbing outer cotton fibrils**	**Turn inside-out + Cold Water (≤85°F) + mild dye-free liquid detergent**
Wavy "Bacon" Ribbed Crewneck Collar	Hanging wet heavy shirt on thin wire hanger or snagged in heavy denim load	Dry flat or drape over drying rack bar; insert hanger from bottom hem only

Source & Complete Care Manual: tshirtsiwant.com/blog/how-to-wash-graphic-tees/

Fit & Shrinkage Tool: tshirtsiwant.com/tools/size-calculator/

APPAREL FABRIC ENGINEERING MATRIX #4 · GSM & DRAPE

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Heavyweight Cotton vs. Ring-Spun vs. Performance Tees Matrix

Side-by-Side Comparison of Fabric Weight (oz/yd² & GSM), Breathability, Sweat Wicking, DTG Print Crispness & Body Drape

1 COMPLETE 4-FABRIC TECHNICAL SPECIFICATION MATRIX

Specification / Metric	Heavyweight Carded Cotton (e.g., Gildan 5000)	Combed Ring-Spun Cotton (e.g., Bella+Canvas 3001)	50/50 Cotton-Poly Blend (Workwear / Crew Tee)	100% Micro-Poly Performance (Moisture-Wicking Athletic)
Fabric Weight (US / Metric)	**5.3 – 6.0 oz/yd²** (180–203 GSM)	**4.2 – 4.5 oz/yd²** (142–153 GSM)	**5.0 – 5.5 oz/yd²** (170–186 GSM)	**3.5 – 4.0 oz/yd²** (118–135 GSM)
Yarn Count & Hand-Feel	**18–20 singles open-end; structured, rugged feel**	**30–32 singles combed; ultra-soft, smooth surface**	**20–26 singles blended; durable, wrinkle-resistant**	**Filament micro-denier knit; slick, cooling touch**
Dry-Heat Breathability	★★★★☆ (High natural airflow)	★★★★★ (Lightweight & airy)	★★★★☆ (Balanced airflow)	★★★★☆ (Engineered mesh pores)
Heavy Sweat / Moisture Wicking	★★☆☆☆ (Absorbs up to 25% weight)	★★☆☆☆ (Holds moisture when soaked)	★★★☆☆ (Dries 40% faster than pure cotton)	★★★★★ (Hydrophobic capillary wicking)
DTG Graphic Print Crispness	**★★★★☆ (Bold opaque ink coverage)**	**★★★★★ (Tightest weave = sharpest detail)**	**★★★★☆ (Slight vintage heathered look)**	**★★☆☆☆ (Requires sublimation or special poly ink)**
Odor Resistance (Multi-Hour Wear)	★★★★★ (Natural cellulose resists odor)	★★★★★ (Stays fresh all day)	★★★★☆ (Good everyday odor control)	★★☆☆☆ (Oleophilic poly traps body oils/bacteria)
Silhouette & Midsection Drape	Boxy tubular cut; hides midsection best	Retail side-seamed; contoured chest & sleeves	Classic workwear cut; zero cling	Athletic drape; tends to cling if sizing is snug

2 FABRIC WEIGHT GUIDE (OZ/YD² TO GSM)

Category	Weight	Best Climate / Use
Featherlight	3.5–4.0 oz (120–135 GSM)	90°F+ humid workouts & running
Retail Mid-Light	**4.2–4.5 oz (142–155 GSM)**	**Everyday casual & layering**
Classic Heavy	**5.3–6.0 oz (180–203 GSM)**	**All-season graphic tees & workwear**
Streetwear Boxy	6.5–8.5 oz (220–288 GSM)	Cool weather & structured streetwear

3 WHICH SHIRT SHOULD YOU CHOOSE?

- **Everyday Graphic Tees & Gifts:** Pick **5.3 oz Classic Heavy Cotton** (durable, hides torso lines) or **4.2 oz Combed Ring-Spun** (softest retail drape).
- **Outdoor Trade Crews (Mechanics / Truckers):** Pick **50/50 Cotton-Poly** or **5.3 oz Cotton**.
- **High-Sweat Gym / 5K Races:** Pick **Moisture-Wicking Performance** (size up 1 step if between sizes to prevent cling).

Sizing Tip Between Fabrics: Heavyweight 5.3 oz tubular cotton (Gildan 5000) hangs straight from the shoulders, while 4.2 oz ring-spun tees and athletic performance shirts drape closer to the body. Use our interactive calculator below to compare exact chest ease across cuts.

Source & Full Fabric Guide: tshirtsiwant.com/blog/cotton-vs-performance-shirts/

Compare Fit by Height/Weight: tshirtsiwant.com/tools/size-calculator/